COMP5566 Project Presentation

Quantifying Frontrunning Activities on Ethereum and Their Impacts on Marketplace

Group 1

1. Project Brief & Requirements

Goal: Collect historical Ethereum DEX transactions, detect frontrunning patterns, quantify adversary profits, and measure market impact on Uniswap V2.

Project 8 — Main steps

1. Collect historical Ethereum transactions
2. Detect Frontrunning activities (Displacement, Insertion, Suppression)
3. Analyze the profits of adversaries
4. Analyze the impact on Ethereum marketplace (e.g., token price)

Our pipeline

```
Etherscan API v2  →  Event Decoder  →  CSV Storage
    →  4 MEV Detectors  →  Profit Analyzer  →  Charts + Excel
```

2. Requirement Coverage

Lecture Requirement	Our Implementation
Collect historical transactions	Etherscan API v2, 5 Uniswap V2 pairs, adaptive block chunking, checkpoint resume
Detect Displacement	Same-block, same-direction, gas ratio >= 1.5x
Detect Insertion	Sandwich detector: front-run + victim + back-run in same block
Detect Suppression	Entity >= 3 swaps with >= 3x block median gas
Detect Arbitrage / Back-running	Opposite-direction trade after large swap (>P90), within 3 tx positions
Analyze adversary profits	USD profit model for sandwich & arbitrage, gas cost estimation, dynamic BTC/ETH pricing
Analyze market impact	Gas price comparison, price time-series, block/volume penetration

3. Data Collection & Preprocessing

5 Uniswap V2 pairs: WETH/USDC, WETH/USDT, WETH/DAI, WETH/WBTC, USDC/USDT
Events: `Swap`, `Sync`, `Mint`, `Burn` from block `10000835` to chain tip

Key engineering

- Adaptive block windows (25k default, auto shrink/grow)
- Multi-key rate limiter with daily quota tracking
- Checkpoint resume + parallel fetching

Preprocessing

- **Entity ID**: if sender is UniswapV2 router, use `to` address instead
- **Ordering**: sort by `(block_number, tx_index, log_index)`
- **ETH price index**: built from WETH/USDC reserves (fallback USDT, DAI)

4. Detection Methods

Type	Logic	Key Threshold
Sandwich (Insertion)	Same entity, same block, opposite-direction pair with victim swap between them	victim same direction as front-run
Displacement	Same block, same direction, different entities, frontrunner executes first	gas ratio >= **1.5x**
Arbitrage / Back-run	Opposite-direction trade immediately after a large swap (>P90)	within **3 tx** positions
Suppression	Single entity >= 3 swaps in block with extreme gas	gas premium >= **3x** block median

Profit model (sandwich):

$$\mathrm{Net} = \mathrm{USD}(n_0) + \mathrm{USD}(n_1) - \frac{(g_f + g_b) \times 150000}{10^{18}} \times P_{ETH}$$

5. Overall Detection Results

Type	Events	Key Metric	Proportion
Sandwich	845 (16.2%)	Net profit: **$194,985**	21.2% profitable
Displacement	758 (14.5%)	Avg gas ratio: **2,258x**	ordering advantage
Arbitrage / Back-run	3,593 (68.9%)	Net profit: **$9,895,544**	avg $2,754/event
Suppression	18 (0.3%)	Avg gas premium: **907x**	extreme but rare
Total	**5,214**	**$10.1M+** aggregate	

- Arbitrage / back-running dominates both in **frequency** and **total profit**
- Sandwich profitability is **highly concentrated** — most attempts lose money on gas
- Suppression is the rarest but shows **extreme** gas competition (907x block median)

6. Sandwich Attack Findings

- **845** events, only **179** profitable (**21.2%**) — profitability highly concentrated
- Gross profit: **$668,920**; Net profit (after gas): **$194,985**
- Avg net: $231; Median net: **-$23** (most attackers lose money on gas)

7. Displacement, Arbitrage

Displacement (758 events) — ordering advantage via gas premium

- Avg gas ratio: **2,258x** (extreme outliers); Top frontrunner: **82** events

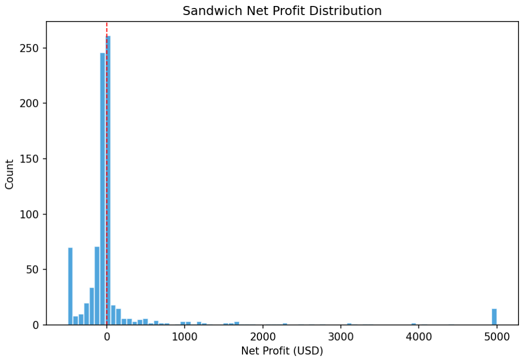
Arbitrage / Back-running (3,593 events) — most frequent pattern

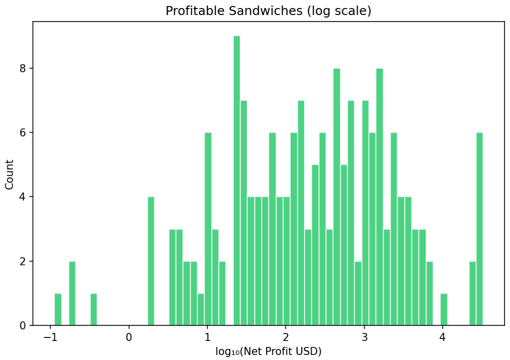
- Net profit: **$9,895,544** (avg **$2,754** per event); Top back-runner: **210** events

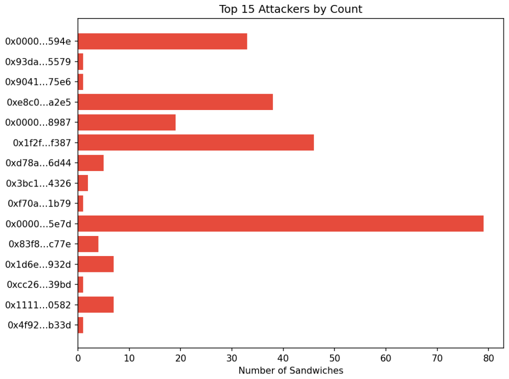
Suppression (18 events) — rarest but most extreme, avg gas premium: **907x**

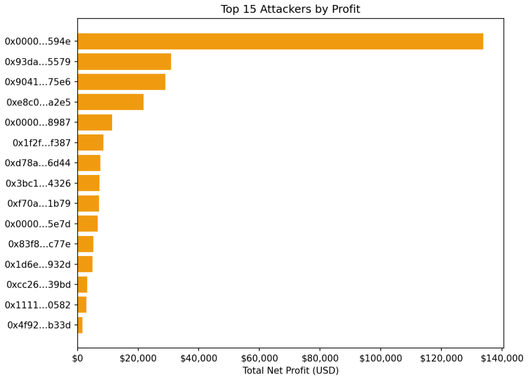
8. Profit Distribution & Top Attackers

- Profit is **extremely concentrated** — a power-law distribution
- Top **10** back-run events account for **85.1%** of total arbitrage profit ($8.4M / $9.9M)
- Top **50** events account for **94.3%** — the remaining **3,543** events share only **5.7%**









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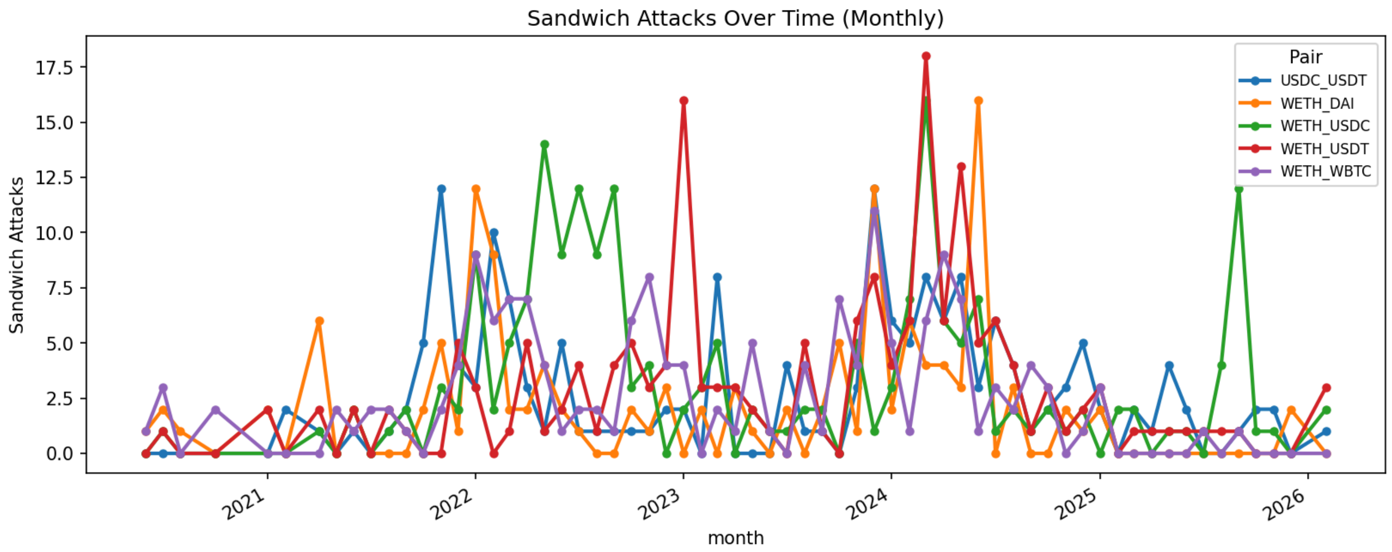
9. Multi-Strategy Entities & Timeline

5 entities operate across **all 4 attack types** simultaneously — evidence of sophisticated MEV bots

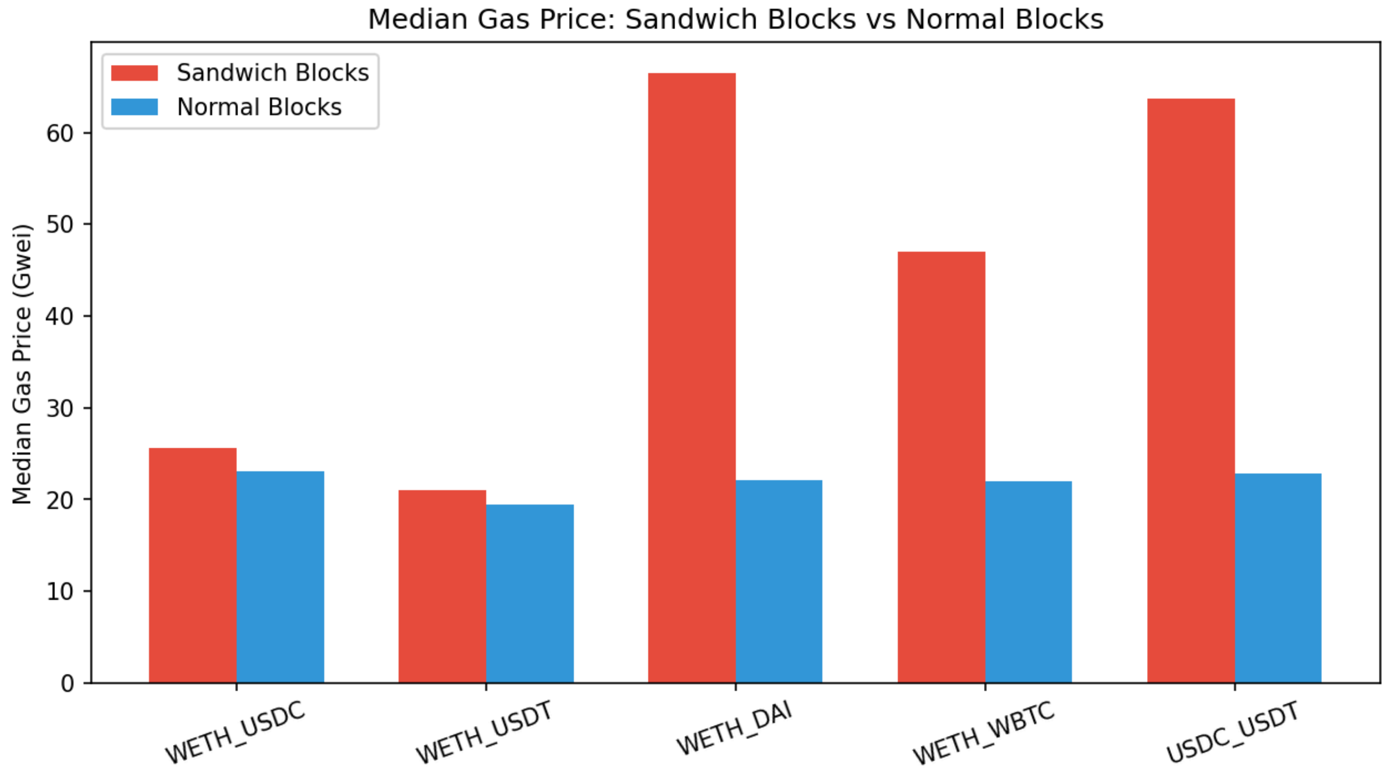
Entity	Strategies
`0x00000000…120e49`	Sandwich + Displacement + Arbitrage + Suppression
`0x6b75d8af…009a80`	Sandwich + Displacement + Arbitrage + Suppression
`0x3328f7f4…309c49`	Sandwich + Displacement + Arbitrage + Suppression

Sandwich timeline: peak activity in **March 2024** (52 events/month), declining afterward as Flashbots/MEV-Boost shifted MEV extraction off-chain



10. Market impact

- Sandwich-related blocks show **significantly higher** median gas prices
- **WETH_DAI**, **WETH_WBTC**, **USDC_USDT**: large gas premiums in suspicious blocks
- MEV activity increases gas costs for **all users** in affected blocks
- Entity `0x6b75…80` appears in both back-running and suppression, suggesting **multi-strategy** operation



11. Limitations

- **On-chain only**: no mempool visibility; dropped/replaced txs invisible
- **Heuristic entity ID**: router-based approach may merge/split entities
- **Fixed gas estimate**: 150k gas per swap is an approximation
- **Price approximation**: reserve-derived ETH prices, dynamic BTC/ETH ratio from pool reserves
- **Threshold sensitivity**: 1.5x, P90, 3x thresholds affect detection counts
- **Interpretation**: patterns are *consistent with* MEV but do not prove intent

12. Project Summary

- Complete pipeline: **collection -> detection -> profit analysis -> visualization**
- Covers all **Project 8 requirements**: Displacement, Insertion, Suppression + bonus Arbitrage
- **5,214 events** detected, **$10.1M+** aggregate adversary profit quantified
- Gas-price analysis confirms MEV raises costs for normal traders

Deliverables

- Python source code (fetcher + analysis pipeline)
- Collected datasets (CSV) + 9 analysis charts + Excel workbook
- Markdown report + GitHub repo for reproducibility

Thank You